

Trigonometric functions and equations - Set 1 - Solutions

Specification MM4.3-MM4.6.

Answer key

Question	Answer	Topic
1	C	Trigonometric functions and equations
2	D	Trigonometric functions and equations
3	D	Trigonometric functions and equations
4	A	Trigonometric functions and equations
5	E	Trigonometric functions and equations
6	B	Trigonometric functions and equations
7	B	Trigonometric functions and equations
8	E	Trigonometric functions and equations
9	B	Trigonometric functions and equations
10	D	Trigonometric functions and equations
11	B	Trigonometric functions and equations
12	C	Trigonometric functions and equations

Worked solutions

Question 1 TRG-01 · Trigonometric functions and equations · spec MM4.6

How many solutions does $\sin 2x = \frac{1}{2}$ have for $0 \leq x < 2\pi$?

Answer: C

$2x$ runs over $[0, 4\pi)$, which contains two full periods of $\sin$, each giving two solutions: 4 in total.

Question 2 TRG-03 · Trigonometric functions and equations · spec MM4.3

Find the exact value of $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$.

Answer: D

$$\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4} + \frac{1}{4} = 1.$$

Question 3 TRG-04 · Trigonometric functions and equations · spec MM4.6

Find the sum of all solutions of $\tan x = \sqrt{3}$ in the interval $0 \leq x \leq 2\pi$.

Answer: D

$$x = \frac{\pi}{3} \text{ and } x = \frac{\pi}{3} + \pi = \frac{4\pi}{3}. \text{ Sum } \frac{5\pi}{3}.$$

Question 4 TRG-06 · Trigonometric functions and equations · spec MM4.5

Given that $\sin \theta = \frac{3}{5}$ and θ is obtuse, find $\tan \theta$.

Answer: A

$$\cos \theta = -\frac{4}{5} \text{ (negative in the second quadrant), so } \tan \theta = \frac{3/5}{-4/5} = -\frac{3}{4}.$$

Question 5 TRG-07 · Trigonometric functions and equations · spec MM4.6

Solve $\sin^2 x = \frac{3}{4}$ for $0 \leq x < 2\pi$. How many solutions are there?

Answer: E

$$\sin x = \pm \frac{\sqrt{3}}{2}: x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}. \text{ Four solutions.}$$

Question 6 TRG-08 · Trigonometric functions and equations · spec MM4.6

Find the smallest positive solution of $\cos\left(x - \frac{\pi}{6}\right) = -\frac{1}{2}$.

Answer: B

$$x - \frac{\pi}{6} = \frac{2\pi}{3} \text{ gives } x = \frac{5\pi}{6} \text{ (the other solution } x - \frac{\pi}{6} = \frac{4\pi}{3} \text{ gives } \frac{3\pi}{2}, \text{ larger).}$$

Question 7 TRG-09 · Trigonometric functions and equations · spec MM4.5

Simplify $\frac{1 - \cos^2 \theta}{\cos \theta \tan \theta}$ for values of θ where the expression is defined.

Answer: B

$$\cos \theta \tan \theta = \sin \theta, \text{ and } 1 - \cos^2 \theta = \sin^2 \theta. \text{ So the expression is } \sin \theta.$$

Question 8 TRG-11 · Trigonometric functions and equations · spec MM4.4

The function $f(x) = 3\sin(2x) - 1$ is defined for all real x . Find the difference between its maximum and minimum values.

Answer: E

Range is $[-4, 2]$; the difference is 6.

Question 9 TRG-12 · Trigonometric functions and equations · spec MM4.6

Find the number of solutions of $\tan^2 x = 3$ for $0 \leq x < 2\pi$.

Answer: B

$\tan x = \pm\sqrt{3}$: $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$. Four solutions.

Question 10 TRG-13 · Trigonometric functions and equations · spec MM4.5

For all θ where the expressions are defined, $(\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^2$ equals

Answer: D

Expanding: $(1 + 2\sin \theta \cos \theta) + (1 - 2\sin \theta \cos \theta) = 2$.

Question 11 TRG-16 · Trigonometric functions and equations · spec MM4.6

Given that $\sin x + \cos x = 0$ and $0 \leq x < 2\pi$, find the sum of all possible values of x .

Answer: B

$\tan x = -1$: $x = \frac{3\pi}{4}$ or $\frac{7\pi}{4}$. Sum = $\frac{10\pi}{4} = \frac{5\pi}{2}$.

Question 12 TRG-18 · Trigonometric functions and equations · spec MM4.3

Find the value of $\tan 30^\circ \tan 60^\circ + \sin^2 45^\circ$.

Answer: C

$\tan 30^\circ \tan 60^\circ = \frac{1}{\sqrt{3}} \cdot \sqrt{3} = 1$, $\sin^2 45^\circ = \frac{1}{2}$. Total $\frac{3}{2}$.